

LESSON 2-5 • HOMEWORK PACKET • SHEET 1 OF 2**Range and IQR**

Maryland Standard 6.SP.B.5c • Learning target – I can find the range and the interquartile range of a data set.

REMEMBER

- **Range** = maximum – minimum. It is the spread of all the data.
- **IQR** (interquartile range) = $Q3 - Q1$. It is the spread of the middle half.
- One extreme value changes the range a lot but the IQR very little.

WORKED EXAMPLE

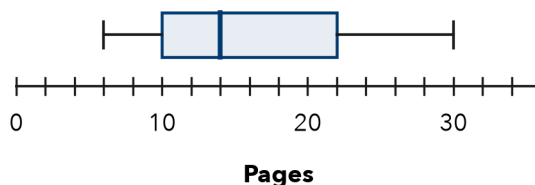
Find the range and IQR of 3, 7, 8, 10, 12, 15, 20.

Step 1 Range: $20 - 3 = 17$ **Step 2** Median 10. $Q1 = 7$ (of 3, 7, 8). $Q3 = 15$ (of 12, 15, 20).**Step 3** IQR: $15 - 7 = 8$ **Answer** Range 17, IQR 8**PRACTICE***show your work • label units*

1. Find each measure.

a. Range of 14, 6, 21, 9: _____**b.** Range of 52, 38, 47, 60: _____**c.** IQR when $Q1 = 18$, $Q3 = 31$: _____**d.** IQR when $Q1 = 40$, $Q3 = 56$: _____**e.** Range of 75, 90, 68, 83: _____**f.** IQR when $Q1 = 7$, $Q3 = 25$: _____

- 2.
- Pages Read.**
- The box plot shows pages read by students in one week.

a. Range: _____ pages**b.** IQR: _____ pages

- 3.
- Jump Rope.**
- Seven students jumped rope 35, 52, 41, 60, 28, 47, and 55 times. Find the range and the IQR.

a. Range: _____**b.** IQR: _____*Show your work.*

- 4.
- Work Backward.**
- A data set has a minimum of 14 and a range of 29. Its
- $Q1$
- is 20 and its IQR is 11.

a. Maximum: _____**b.** $Q3$: _____

- 5.
- Find the Error.**
- Mateo says the IQR is always half of the range. Use the Jump Rope data to explain why he is wrong.

CHECK YOURSELF☐ I can do this on my own☐ I need a little help☐ I have a question about #_____**AT HOME** Ask family members their ages. Find the range of the ages.

LESSON 2-5 • HOMEWORK PACKET • SHEET 2 OF 2

Comparing Variability

Maryland Standard 6.SP.B.5c • Learning target – I can compare the variability of two data sets using range and IQR.

REMEMBER

- More range or IQR = **more variability**.
- Less variability = more **consistent**. Compare range with range, IQR with IQR.
- With an extreme value, use the **IQR**.

WORKED EXAMPLE

Team P: IQR 6 points. Team Q: IQR 15 points. Which is more consistent?

Step 1 $6 < 15$, so Team P's middle half is closer.**Answer Team P****PRACTICE***show your work • label units*

1. Find each IQR.

a. Set P: $Q1 = 30$, $Q3 = 42$ _____**b.** Set Q: $Q1 = 33$, $Q3 = 39$ _____

- 2.
- Basketball.**
- Points per game: Player 1: 8, 10, 11, 12, 13, 14, 16. Player 2: 2, 6, 9, 12, 15, 18, 22.

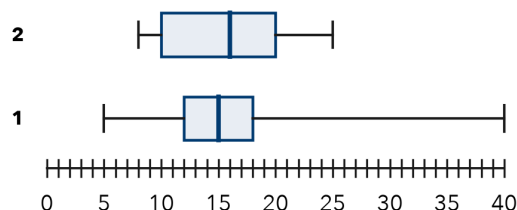
	Range	IQR
Player 1		
Player 2		

More consistent player _____

- 3.
- Spelling Scores.**
- Group A scored 70, 72, 75, 78, 80. Group B scored 55, 65, 75, 85, 95.

Ranges: A _____ B _____ More consistent group: _____

- 4.
- Bus Routes.**
- The box plots show ride times in minutes.

a. Ranges: 1 _____ 2 _____**b.** IQRs: 1 _____ 2 _____

5. Jada says Route 2 is more consistent. Kwame says Route 1 is. Who is right? Why?

TEST PRACTICE*choose the best answer*

6. Class A's scores have a range of 12. Class B's have a range of 30. Whose scores vary less?

A Class A**B** Class B**C** Both the same**D** Cannot tell

7. What is the IQR of 4, 6, 7, 9, 10, 12, 15?

A 6**B** 11**C** 4.5**D** 9**CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about # _____